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Course: SEA4001W

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Chlorophyll variability in the Humboldt Current (specifically between 16-20°S)

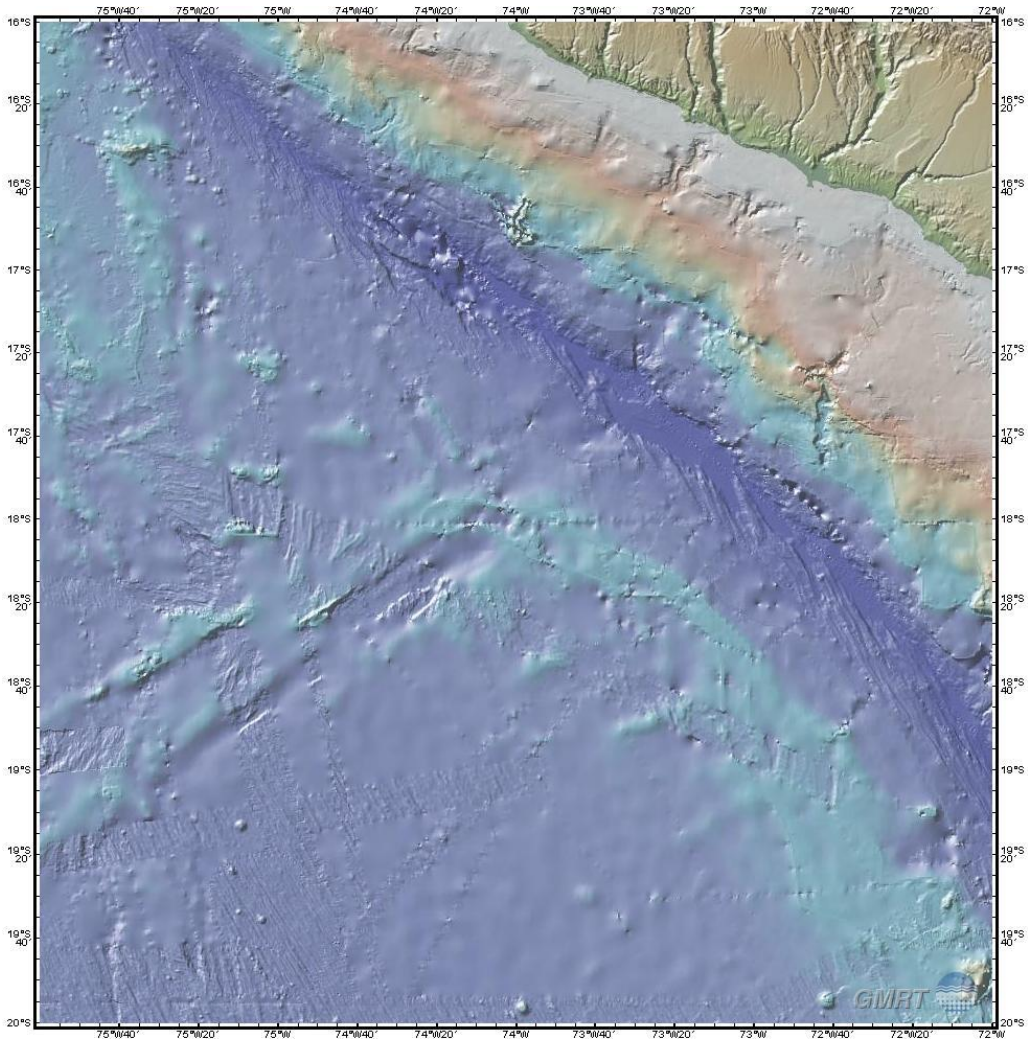

Figure 1: A map that shows a region along the eastern boundary current of South America.

Introduction

The Humboldt Current System (HCS), also known as the Peru current, is an eastern boundary current that runs northward along South America, roughly around 42° S towards the equator. The HCS is highly regarded as being one of the most productive marine ecosystems on the planet and is estimated to support approximately 15% of the global fish catch, making it a cornerstone of global food security (ATA Team, 2026). The region has 3 main coastal upwelling centers (20–22°S, 32-34°S, 36-38°S) (Thiel et al., 2007) so there are large amounts of research focused on these regions looking at how bathymetry, El Nino Southern Oscillation (ENSO) and oxygen affect chlorophyll concentration. After coming to this realization, I was interested in looking at other regions along the HCS that may have contrasting patterns, and I ended up focusing on a region known as the transboundary zone that runs between southern Peru and northern Chile (specifically 16°S-20°S). I found this region to be interesting as productivity is much lower than the surrounding system with upwelling waters confined to the coastal zone (Cunningham et al., 2020).

Data & methods

Chlorophyll-a concentration data were obtained from the ESA Ocean Colour Climate Change Initiative (Ocean_Colour_cci) dataset (Sathyendranath et al., 2023). This dataset provides monthly climatology of the generated ocean colour products covering the period 1997 – 2022 but a smaller set of the data was downloaded that only has data from 1997-1998 (12 months). The dataset is available via a Digital Object Identifier (DOI), ensuring a persistent and citable link to the data source.

The bathymetry data was obtained from the Global Multi Resolution Topography (GMRT) dataset (Ryan et al., 2009) which is a multi-resolution Digital Elevation Model that processes ship-based sonar data that can be gridded into spatial resolution. Specifically the data was downloaded was in the NetCDF format. The dataset is also available via a DOI, ensuring a persistent and citable link to the data source.

Data analysis was conducted using python.

Results

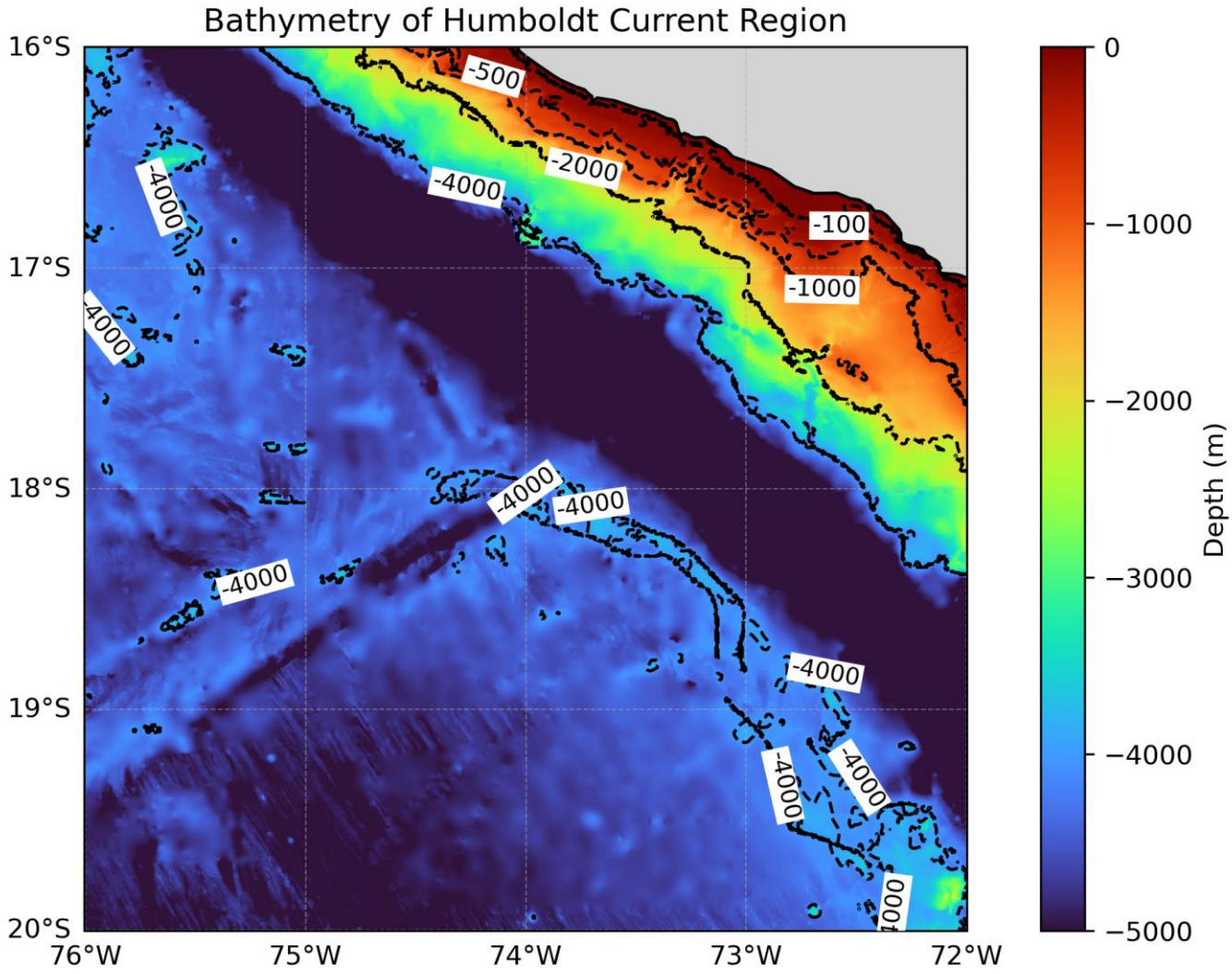

Figure 2: Bathymetry of the Humboldt Current study region. Contours indicate depth (m), showing a narrow continental shelf and steep offshore slope.

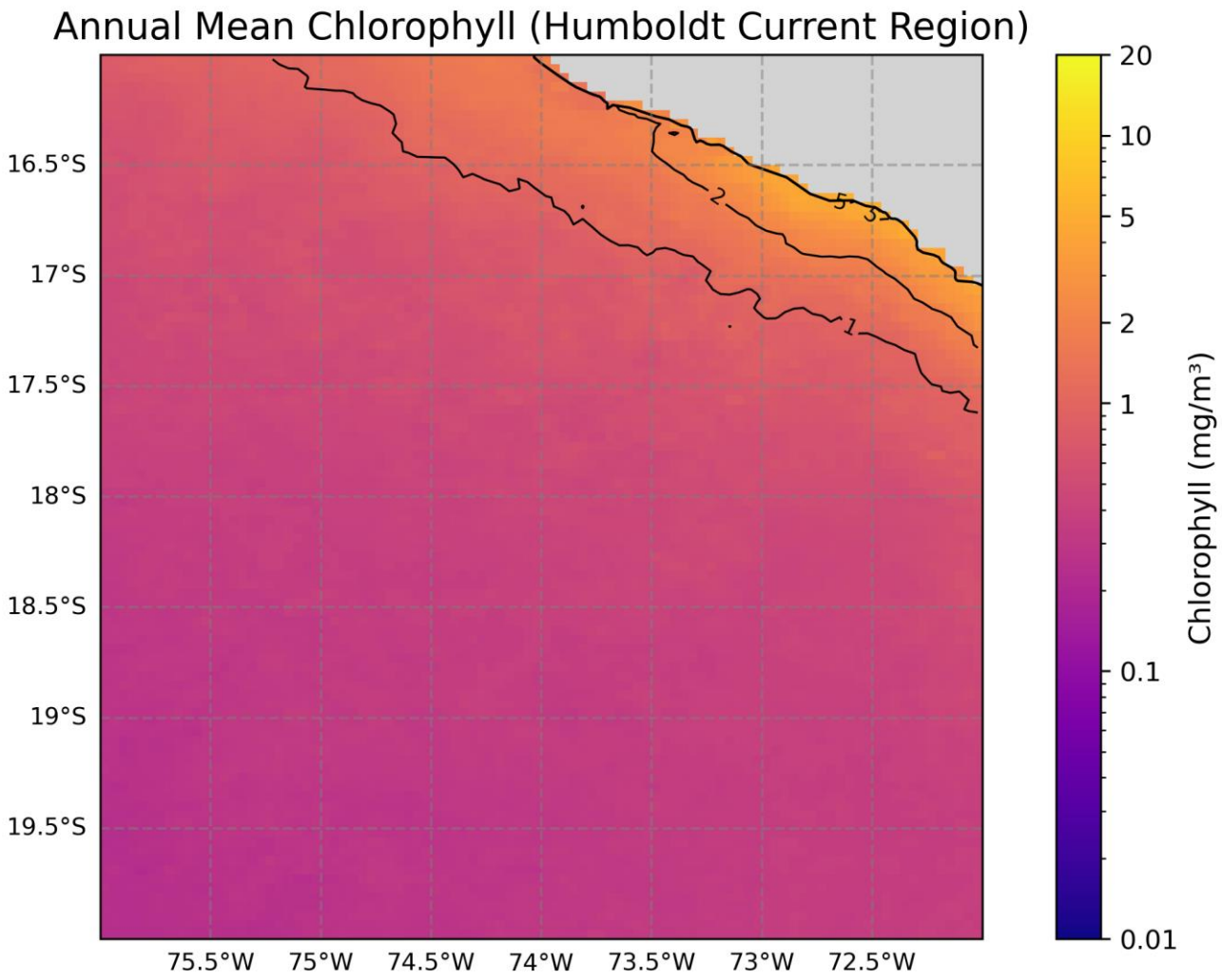

Figure 3: Annual mean chlorophyll concentration (mg.m^{-3}) in the Humboldt Current region. Higher mean chlorophyll concentrations along the coast linked with the upwelling nutrient-rich waters.

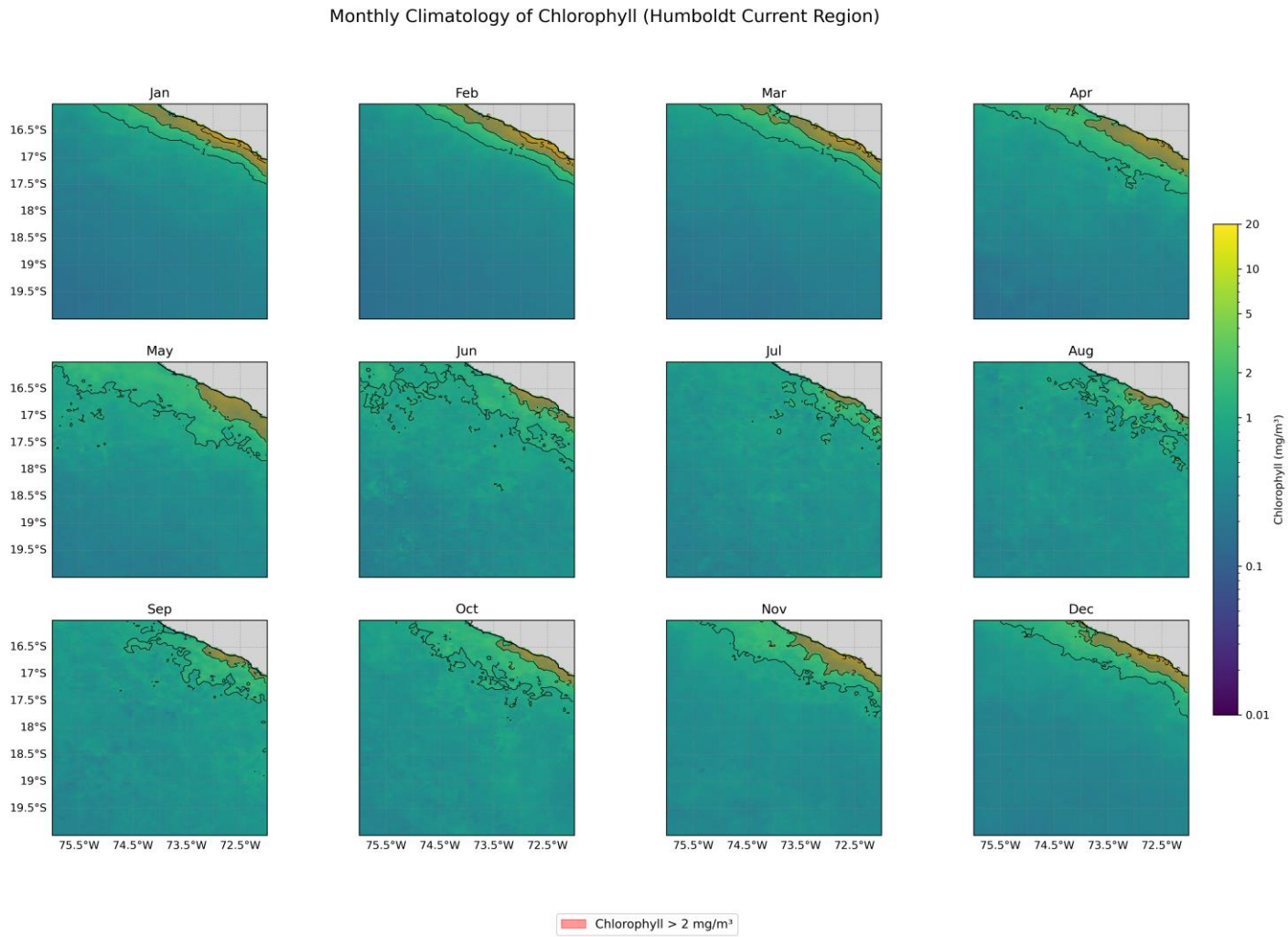

Figure 4: Monthly variation of chlorophyll concentration. Red shading indicates regions where chlorophyll exceeds 2 mg m^{-3}, highlighting areas of high productivity. Seasonal variability is relatively weak, with persistently 'high' chlorophyll near the coast.

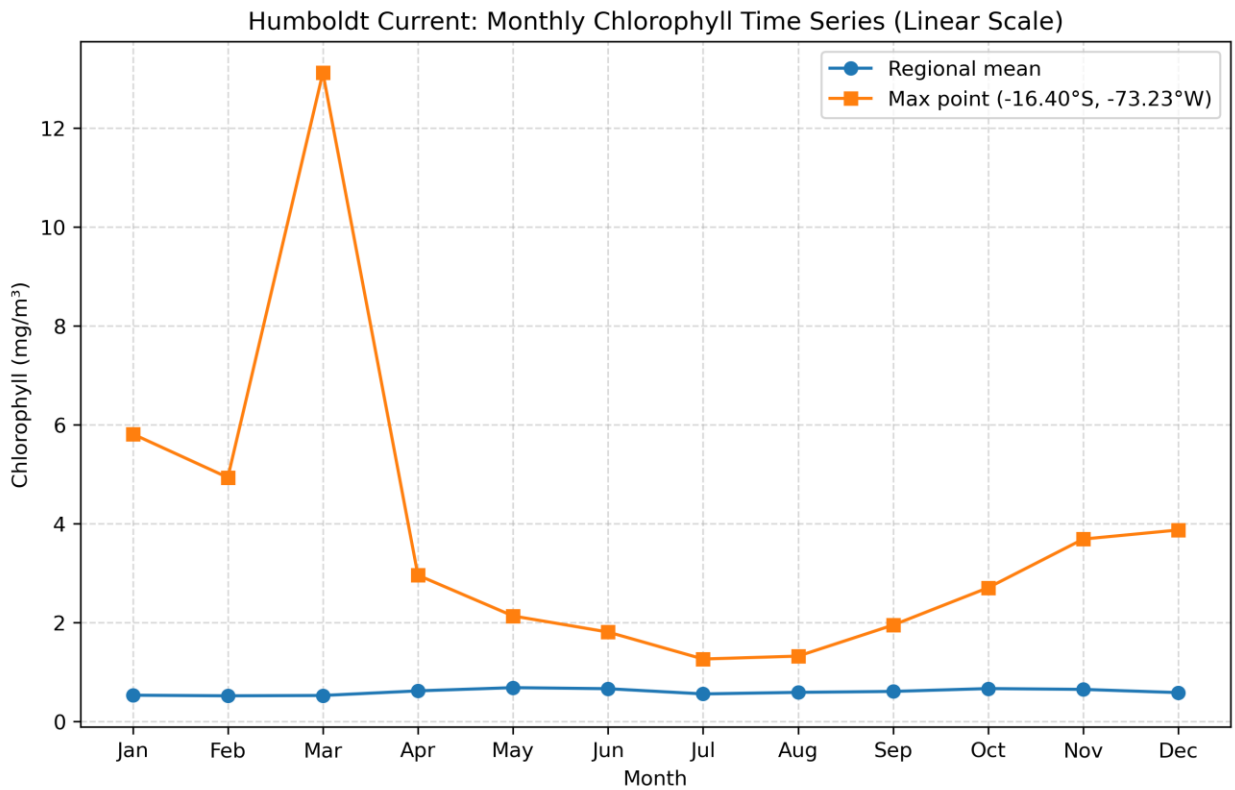

Figure 5: Comparison between the seasonal mean of the entire region and a grid point along an area of high chlorophyll concentration. During summer and spring, the point's concentration is vastly higher than the regional means but as expected during winter their difference is a lot smaller (the seasons are according to the Southern Hemisphere).

Additional plots:

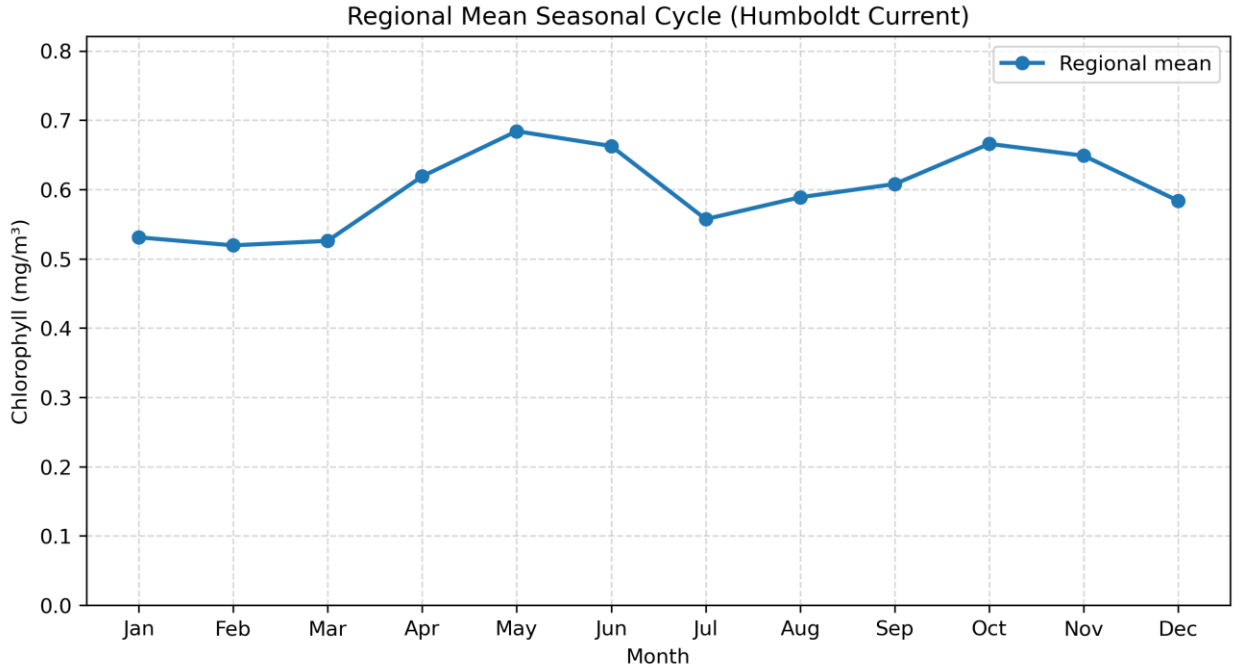

Figure 6: Time series that takes a closer look at the region's monthly chlorophyll means (zoomed in to show the scale clearly), there is a stable trend with the concentration showing that it averages between 0.5-0.7 mg m^{-3}.

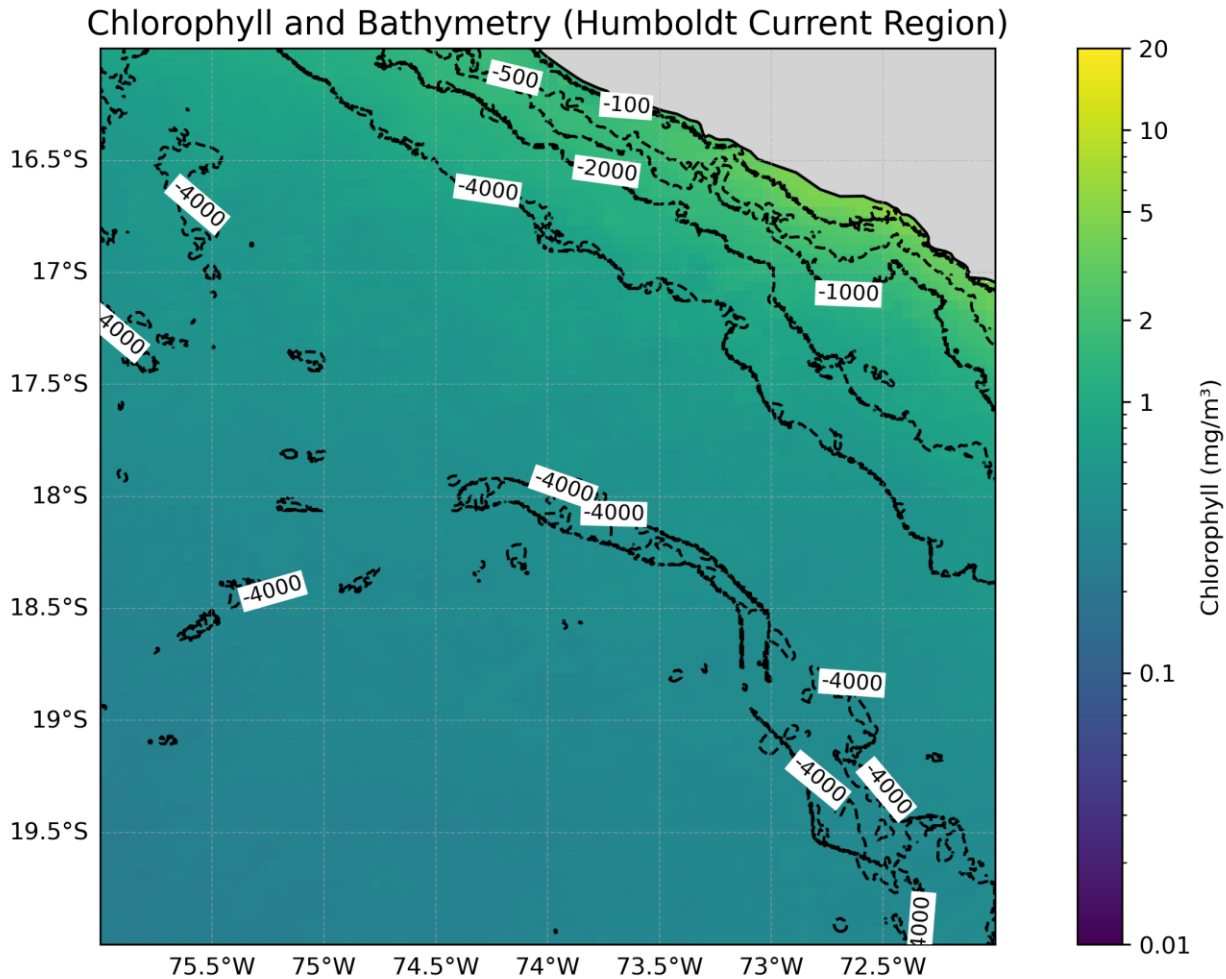

Figure 7: A plot that overlays bathymetry with annual chlorophyll. There is a clear connection between the lower bathymetry regions and higher chlorophyll concentrations.

Discussion

The results from using the data and plotting show the precise bathymetry and the chlorophyll patterns of the region. The bathymetry along the continental shelf shows it is quite narrow and steep, which is a key feature that allows for enhanced upwelling of deeper waters. However as seen in figure 3 the northward coast has consistently low chlorophyll concentration and as you head southwards towards near one of the main upwelling centers the concentration increases significantly (average values around 5 mg m^{-3}). The 12 monthly plots allow for a further look into the seasonality of chlorophyll concentrations and as seen in figure 4 values are relatively high throughout the spring and summer months with peak concentrations present during the February-March period. Through winter there also seems to be some areas that still have concentrations around 2 mg m^{-3} and this shows just how strong the upwelling processes are with light being the limiting factor. Figure 5 reinforces the idea that the regional mean concentration remains stable throughout the year while showing that near the coast there is a strong sesaonal gradient where values are high during summer then drastically lower during the winter-autumn period.

Conclusion

The use of python allowed for the analysis of data to look at and confirm that the transboundary region experiences relatively lower productivity than what is expected of a highly productive current system. This goes to show the complexity around upwelling in eastern boundary currents, and that bathymetry, wind stress, light etc. play an important role when it comes to providing the platform for strong phytoplankton blooms. Further research and data collection would be useful to look at how this transboundary region has maybe changed over time and how outside processes such as ENSO may have impacted this specific dataset (as of the data was collected during 1997-1998 where one of the strongest recorded ENSO's occurred).

Reference list

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